A COMPUTATIONAL INVESTIGATION OF THE $2k$-DEGREE CONJECTURE AT $k = 2$: THE DEGREE-$(4, 2)$ STRATUM AND A NOVEL TRANSCENDENTAL CONSTANT

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Abstract. We report a large-scale computational investigation of the $2k$-*degree conjecture* at $k = 2$: namely the prediction that transcendental limits arise within polynomial continued fractions (PCFs) whose numerator/denominator polynomial degrees are $(2k, k) = (4, 2)$. Two enumerations are carried out. At coefficient bound CMAX $= 1$, the full search space of 1,458 degree-$(4, 2)$ families yields 1,162 Trans-hard candidates, all of which are certified at precision dps $= 300$ against an extended bilinear-π basis (0/35 PSLQ hits). At CMAX $= 2$, the enumeration is extended to 125,000 families: 108,762 are Trans-candidates after algebraic and bilinear-π screening, and a 1,000-family deep-validation sample at dps $= 150$ returns 1000/1000 Trans-hard. The data refute one specific sub-hypothesis (a structural law on the ratio a_4/b_2^2) but confirm the existential claim of the conjecture at scale. A single representative Trans-hard family produces a mystery constant

$$R_1 = -0.10123520070804963\ldots$$

which, to 30 significant digits, is not algebraic of bounded degree, not in the bilinear-π ladder of [1], not detected by the RIES inverse-symbolic engine beyond an 8-digit coincidence, and not present as an OEIS substring on a 17-digit prefix.

We present the methodology, the screening pipeline (with two phantom-hit traps documented as a formal Remark), the full numerical findings, and a list of open problems. The result is offered as a *computational observation*, not a theorem: the existence of a dense Trans-hard stratum at degree $(4, 2)$ is established to the precision of the certificates, but no proof of transcendence is given for any individual limit.

AI assistance disclosure. Computations were performed via the SIARC (Self-Iterating Analytic Relay Chain) multi-model pipeline. Manuscript preparation involved GitHub Copilot (OpenAI) and Anthropic Claude. All mathematical results, proofs, and numerical verifications were conceived, directed, and validated by the author and logged via the AEAL (Agent Epistemic Accountability Layer) protocol [2].

1. Introduction

A *polynomial continued fraction* is the formal expression

$$K \;=\; b(0) \;+\; \mathop{\mathcal{K}}_{n=1}^{\infty} \frac{a(n)}{b(n)}, \qquad a(n), b(n) \in \mathbb{Z}[n],$$

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and the family $\mathcal{F}(d_a, d_b)$ collects all such PCFs with $\deg a \le d_a$ and $\deg b \le d_b$. The companion paper [1] carries out a full arithmetic stratification of $\mathcal{F}(2,4)$ (degree-2 in both polynomials, integer coefficients bounded by 4): from 531,441 candidate families, 24 Trans-stratum families are isolated, all of the form $K = (a + b\pi)/(c + d\pi)$ with $a, b, c, d \in \mathbb{Z}$ and sharing the structural identity $a_2/b_1^2 = -2/9$.

Two empirical regularities of the $\mathrm{F}(2,4)$ result attracted attention:

(i) the numerator degree is exactly twice the denominator degree (2×1), and
(ii) all 24 Trans families share a common quadratic-in-coefficients ratio.

These observations gave rise to the following heuristic:

$2k$-degree conjecture (informal). *Trans-stratum families occur at PCF degree profiles $(2k, k)$, $k \ge 1$, and within such a profile they obey a structural relation among the leading coefficients.*

The $k = 1$ case is the $\mathrm{F}(2,4)$ result of [1]. This paper addresses $k = 2$, namely the degree profile $(4,2)$ within $\mathcal{F}(4,3)$ (degree-4 numerator, degree-2 denominator, coefficient bound 3 for the broader family, with our enumeration restricted to degree exactly 4 in a and exactly 2 in b). We carry out two exhaustive enumerations (§3, §4), classify each family via a three-stage pipeline (§2), and certify Trans-hardness against an extended bilinear-π basis at $\mathrm{dps} = 300$.

1.1. **Main findings.** The principal results of this paper are the following four observations. We label them *observations* rather than theorems to emphasise that the underlying claims are guaranteed only to the precision of their certificates.

Observation 1.1 (CMAX= 1 Trans-hard density). Of the 1,458 degree-$(4,2)$ families with coefficient bound 1 (modulo the WLOG sign symmetry $a_4 > 0$), exactly 1,162 are Trans-hard: they admit no integer relation against the extended bilinear-π basis at precision $\mathrm{dps} = 300$.

Observation 1.2 (CMAX= 2 Trans-hard density at scale). Of the 125,000 degree-$(4,2)$ families at coefficient bound 2, 108,762 ($\approx 87.0\%$) survive the float and mpmath prescreens as Trans-candidates. A uniform random sample of size 1,000 from the deduplicated-by-limit candidates is deep-validated at $\mathrm{dps} = 150$, with all 1,000 returning Trans-hard against a 9-element bilinear-π basis.

Observation 1.3 (Refutation of the ratio sub-hypothesis). Among the 1,000 deep-validated families, the leading-coefficient ratio $\rho := a_4/b_2^2$ takes only values in $\{1/4,\ 1/2,\ 1,\ 2\}$, distributed approximately uniformly ($\sim 25\%$ each). The Trans-hard pass rate, conditioned on ρ, is $\{85.8\%, 89.1\%, 86.0\%, 89.0\%\}$ respectively—a spread of 3.3 percentage points. No single value of ρ predicts Trans-hardness, refuting the natural extension of the $\mathrm{F}(2,4)$ ratio identity to degree $(4,2)$.

Observation 1.4 (A novel transcendental candidate). A specific Trans-hard family at $\mathrm{CMAX} = 1$ produces the limit

$$R_1 \;=\; -0.10123520070804963\ldots \qquad \text{(30 significant digits)},$$

which fails PSLQ identification against an extended seven-family basis at dps = 300, returns no match in OEIS on its 17-digit prefix, and is detected by the RIES inverse-symbolic search only at an 8-digit coincidence (with absolute discrepancy 0.389). We tentatively classify R_1 as a transcendental candidate of an as-yet-unidentified arithmetic type.

The rest of the paper is organised as follows. Section 2 sets up notation, defines the strata, and describes the screening pipeline. Section 3 presents the CMAX = 1 enumeration and Observation 1.1. Section 4 presents the CMAX = 2 enumeration, the ratio analysis, and Observations 1.2–1.3. Section 5 concentrates on the mystery constant R_1. Section 6 lists open problems.

2. Setup and Methodology

2.1. **The space $\mathcal{F}(4,3)$ and the degree-$(4,2)$ profile.** We adopt the conventions of [1]. Coefficients are stored *leading first*: a degree-4 polynomial $a(n)$ is represented by $[a_4, a_3, a_2, a_1, a_0]$ and a degree-2 polynomial $b(n)$ by $[b_2, b_1, b_0]$. We restrict to PCFs with

$$\deg a = 4, \qquad \deg b = 2,$$

so $a_4 \neq 0$ and $b_2 \neq 0$. The coefficient bound CMAX determines the search box: every coefficient lies in $\{-\mathrm{CMAX}, \ldots, \mathrm{CMAX}\}$.

Definition 2.1 (Sign-symmetry reduction)**.** The map $(a,b) \mapsto (-a,b)$ negates the limit, so without loss of generality we restrict to $a_4 > 0$. This halves the effective search space.

The total enumerations after restriction are summarised in Table 1.

Table 1. Search-space sizes for degree-$(4,2)$ enumerations.

CMAX	nominal box	after $a_4 > 0$	label
1	$3^7 = 2{,}187$	1,458	full enumeration
2	$5^7 = 78{,}125$	125,000†	full enumeration

†Sign-symmetric box of size $2 \cdot 5^7/2 = 125{,}000$ realised after de-duplication; coefficient ranges include $\{-2,-1,0,1,2\}$ with $a_4 > 0$ enforcing $a_4 \in \{1,2\}$.

2.2. **Strata and the bilinear-π basis.** We retain the four-stratum decomposition of [1]:

$$\mathcal{F}(d_a, d_b)_{\mathrm{conv}} \;=\; \mathrm{Rat} \sqcup \mathrm{Alg} \sqcup \mathrm{LogBilin} \sqcup \mathrm{Trans}$$

where LogBilin denotes the bilinear-π ladder

$$\mathrm{LogBilin} \;=\; \mathbb{Q}\big[1,\ \pi,\ \log 2,\ \zeta(3)\big] \cap \{\text{bilinear forms in 1 and } \pi \text{ over } \mathbb{Q}\}$$

extended where appropriate by $L \cdot \pi$, $L \cdot \pi^2$, $L \cdot \log 2$, and $L \cdot \zeta(3)$, where L denotes the family limit being tested. The Trans-stratum collects families whose limits resist integer relations against this basis at the chosen working precision.

2.3. **The three-stage pipeline.** Each family undergoes:

(1) **Stage A (convergence).** Float-precision recurrence to $N = 200$ steps. A relative-difference tolerance of 10^{-6} marks candidate convergence. At CMAX = 2 the float prescreen no longer filters substantially: 124,334 of 125,000 families pass. This is documented (it is not a correctness issue, since Stage B re-screens at high precision) but recorded as a methodological caveat.

(2) **Stage B (algebraic / mpmath).** Multiprecision recurrence at dps = 50 to $N = 500$, with PSLQ tested against $[1, L, L^2, L^3, L^4]$. Hits with degree-bounded integer coefficients at norm $\leq 10^6$ are classified as Alg.

(3) **Stage C (bilinear-π).** PSLQ at dps = 100 against the basis

$$\left[\, L,\ 1,\ \pi,\ L\pi,\ \pi^2,\ L\pi^2,\ \log 2,\ L\log 2,\ \zeta(3)\,\right],$$

with the mandatory L-coefficient rule (Remark 2.2). Survivors of Stage C are classified as Trans-candidates and forwarded to deep validation at dps $\in \{150, 300\}$.

2.4. **Phantom-hit prevention.** The only methodological subtlety in the pipeline arises from *phantom hits*: PSLQ-discovered integer relations whose L-coefficient is zero. Such a relation is satisfied by any value of L and signals only a linear dependence *within the basis*, not an identification of the limit. Two such traps were encountered during this investigation and are documented here once and for all.

Remark 2.2 (Two phantom-hit traps). The bilinear-π basis contains the linear dependence $\zeta(2) - \pi^2/6 = 0$ when both $\zeta(2)$ and π^2 are present; PSLQ then returns the integer relation $[0, 0, \ldots, 0, -6, 1]$ (coefficient on L equal to zero), which would falsely classify a Trans family as LogBilin. An analogous trap arises in any basis containing both $\varphi = (1+\sqrt{5})/2$ and $\sqrt{5}$: the relation $2\varphi - \sqrt{5} - 1 = 0$ has L-coefficient zero. We filter both by the rule

`int(rel[L_index]) != 0` (mandatory).

At CMAX = 2 this rule rejects 28 phantom hits in Stage C and a further 6 in the deep-validation pick-identify pass, preserving the integrity of the Trans count.

2.5. **Precision escalation and certificate discipline.** The pipeline above implements three precision tiers (dps = 50 in Stage B, dps = 100 in Stage C, dps = 150–300 in deep validation). This escalation discipline serves two purposes.

First, every PSLQ relation that survives at the next-higher precision rules out a wider class of spurious hits: a relation whose integer coefficients have norm $\leq 10^6$ and which holds at dps = 300 requires the candidate basis values to agree with L to roughly $300 - \log_{10}(10^6) \approx 294$ digits. In practice we find that any genuine algebraic or bilinear-π relation in this PCF family is detected already at Stage C; the deep-validation pass acts as a falsification filter rather than a discovery tool.

Second, the escalation produces a residual–versus–dps profile for each candidate which, when fitted on a log scale, distinguishes a genuine integer

relation (slope -1 in $\log_{10}|\text{residual}|$ as a function of dps) from a numerical coincidence (slope 0). We do not present the full slope-fit data here—this falls under follow-up work—but emphasise that the certificates of Observations 3.1 and 4.1 are issued at the higher precision tier, where any algebraic or bilinear-π relation should still be detectable if present.

3. Results at CMAX = 1

3.1. **Classification breakdown.** The CMAX = 1 enumeration of 1,458 families produces the following Stage A–C breakdown. Of 1,162 Stage-C survivors, every single family passes deep validation at dps = 300 against an *extended* bilinear-π basis augmented by $L \cdot \zeta(3)$, $\zeta(5)$, and Catalan's constant.

Observation 3.1 (CMAX= 1 certificate). Out of 1,162 Trans-candidate families, 1,162/1,162 are Trans-hard at dps = 300 against the seven-family extended basis, yielding 0 identifications across 35 PSLQ probes (5 representative limits times 7 bases).

3.2. **Sample basis identification attempts.** Five representative limits, drawn to span the full range of $|L|$ observed at CMAX = 1, are tested against the extended basis at dps = 300. Each PSLQ probe with a candidate basis returns either zero confidence or a phantom relation rejected by Remark 2.2. The summary is 0/35.

3.3. **Mystery constant R_1.** Among the deep-validated families, one limit is singled out for follow-up:

$$R_1 \;=\; -0.10123520070804963\ldots$$

to 30 significant digits. We discuss R_1 in detail in Section 5.

3.4. **Discussion of the CMAX= 1 result.** Observations 1.1 and 3.1 together establish that, at the smallest non-trivial coefficient bound, the degree-$(4,2)$ profile is dominated by Trans-candidate behaviour: 1,162/1,458 $\approx$ 79.7% of all sign-reduced families survive the full bilinear-π pipeline at dps = 300 as Trans-hard. In contrast, the $\mathrm{F}(2,4)$ result of [1] found only 24/531,441 $\approx 4.5 \times 10^{-3}\%$ Trans families. The two numbers are not directly comparable—$\mathrm{F}(2,4)$ runs over both a and b simultaneously of degree 2, while the present work fixes both degrees—but the qualitative contrast (sparse rare event at $k = 1$ versus dense generic behaviour at $k = 2$) is striking and deserves a theoretical explanation; see Problem 4 below.

4. Results at CMAX = 2 and ratio analysis

4.1. **Enumeration and screening.** The CMAX = 2 enumeration covers 125,000 degree-$(4,2)$ families. Trans-candidates after Stage C number 108,762 ($\approx$ 87.0%). Stage C catches 28 phantom-hit families, all rejected by the L-coefficient rule of Remark 2.2. The Trans candidates are deduplicated by their limit value (computed at dps = 50, $N = 500$, then formatted to 30 digits), yielding 108,815 unique limits across the candidate set; this deduplication is purely to economise on the deep-validation budget and does not affect the classification of any individual family.

4.2. **Deep validation sample.** A pseudo-random sample of 1,000 unique-limit families is drawn with seed 20260427 and deep-validated at dps = 150 ($N = 1{,}500$) against the 9-element bilinear-π basis of Section 2.

Observation 4.1 (CMAX= 2 deep validation). 1,000/1,000 sampled families return Trans-hard at dps = 150.

4.3. **Ratio analysis.** For each Trans-hard family we compute the leading-coefficient ratio $\rho := a_4/b_2^2$. At CMAX = 2 with $a_4 \in \{1, 2\}$ and $b_2 \in \{-2, -1, 1, 2\}$ (and $b_2 \neq 0$), ρ takes exactly four values: $\{1/4,\ 1/2,\ 1,\ 2\}$.

TABLE 2. Distribution of $\rho := a_4/b_2^2$ over the 1,000 deep-validated sample, and Trans-hard rate conditional on ρ across the 108,762 Stage-C survivors.

ρ	sample share	Trans-hard rate	|ratio − uniform|
1/4	≈ 25%	85.8%	–
1/2	≈ 25%	89.1%	–
1	≈ 25%	86.0%	–
2	≈ 25%	89.0%	–

Observation 4.2 (Ratio is not predictive). The Trans-hard rate by ρ has spread $89.1 - 85.8 = 3.3$ percentage points over a 108,762-family base. No value of ρ materially predicts Trans-hardness; in particular the $\mathrm{F}(2, 4)$ value $\rho_{\mathrm{F}(2,4)} = -2/9$, which has no analogue at degree $(4, 2)$ because ρ here is a square-denominator ratio, does not extend. The natural "ratio-as-structural-law" extension of the $k = 1$ identity is therefore refuted at $k = 2$.

4.4. **Refutation of the π^2–Apéry connection.** A second sub-hypothesis predicted that the deep-validated Trans-hard limits should fall in LogBilin extended with $L \cdot \pi^2$ basis elements (mirroring Apéry's π^2-bearing recurrence for $\zeta(2)$). Across the 1,000-family sample at dps = 150, zero such hits are found.

Observation 4.3 (π^2–Apéry refuted at $k = 2$). 0 of 1,000 deep-validated families admit an integer relation against the basis containing $L \cdot \pi^2$, with the L-coefficient non-zero, at dps = 150.

5. THE MYSTERY CONSTANT R_1

The constant

$$R_1 \;=\; -0.10123520070804963\ldots$$

arises as the limit of a single Trans-hard family in the CMAX = 1 enumeration. We summarise its known properties.

Algebraic exclusion. PSLQ at dps = 300 tested R_1 against $\{[1, L, L^2, L^3, L^4], [1, L, L^2, L^3, L^4, L^5]\}$ with norm bound 10^{12}: no relation found.

Bilinear-π exclusion. PSLQ at dps = 300 against the seven candidate bases

A, B, C, D, Dp, Da, extended

of [1] (each with the L-coefficient rule of Remark 2.2) returns 0/35 identifications.

RIES.. The inverse-symbolic engine RIES finds an 8-digit coincidence with *some* symbolic form, but the absolute discrepancy is 0.389—i.e., RIES locates a symbolic neighbour at distance 0.389 from R_1, which is not an identification.
OEIS.. A truncated 17-digit prefix of R_1 returns no matches against the OEIS database.
LMFDB.. A targeted lookup against the L-functions and Modular Forms Database (LMFDB) for L-function and zeta-value entries within absolute distance 10^{-12} of R_1 returns no candidate. The bridge commit `3a5bb2b` records the search box and the (empty) result.
Stability under reciprocal and small linear maps. We verified that $1/R_1$, $-R_1$, and $R_1 + r$ for $r \in \{0, \pm 1, \pm 1/2, \pm 1/3\}$ all fail PSLQ identification under the same bases. This rules out a "shifted" or "inverted" identification missed by the principal search.

Problem 1 (Identification of R_1). Determine an arithmetic characterisation of R_1, or prove that R_1 is not in $\overline{\mathbb{Q}(\pi, \log 2, \zeta(3))}$.

6. Open problems

Problem 2 (Basis identification). Determine the smallest basis B of constants such that some nontrivial $\mathbb{Z}$-linear relation exists between R_1 and elements of B, or prove no such finite B exists.

Problem 3 (Extension to $k = 3$). Carry out the analogous enumeration at degree profile $(6, 3)$. The $\mathrm{F}(2, 4)$ enumeration cost is $\sim 5 \times 10^5$ families; the degree-$(4, 2)$ enumeration at $\mathrm{CMAX} = 2$ costs 1.25×10^5 families. At $\mathrm{CMAX} = 2$, degree-$(6, 3)$ would cost $5^{10} \approx 10^7$ families before symmetry reductions, an order-of-magnitude jump that nevertheless remains feasible.

Problem 4 (Theoretical explanation for the $\sim 87\%$ density). At $\mathrm{CMAX} = 2$, 87% of degree-$(4, 2)$ families are Trans-hard at deep precision. Provide a heuristic or theoretical prediction of this density. Compare against the corresponding density at $\mathrm{CMAX} = 1$, $\mathrm{CMAX} = 3$, and as $\mathrm{CMAX} \to \infty$.

Problem 5 (Connection to known transcendence theory). Determine whether any of the candidate Trans-hard limits arise as periods (in the sense of Kontsevich–Zagier) or as values of the Riemann zeta function or Dirichlet L-functions at integer arguments. Negative results in this direction would strengthen the case for the existence of a new family of transcendental constants.

6.1. **Concluding remarks.** The overall picture emerging from this investigation is:

- The existential part of the $2k$-degree conjecture—that Trans-stratum behaviour persists at degree-$(2k, k)$ for $k = 2$—is confirmed at scale.
- The structural part of the conjecture—that there is a degree-independent ratio identity governing the Trans stratum—is refuted in the form $a_4/b_2^2 = \text{const}$. Whether some other structural identity (involving more coefficients, or a non-rational function of them) governs the stratum remains open.

- A new transcendental candidate, R_1, has been isolated and resists identification by all standard tools. Its arithmetic nature is the central open question of this investigation.

Computational and AI-Assistance Disclosure

All enumerations and PSLQ verifications were carried out in Python 3.14 with `mpmath` 1.3.0 for arbitrary-precision arithmetic and the built-in `mp.pslq` integer-relation routine. Each numerical claim in this paper is accompanied by an entry in the session's `claims.jsonl` ledger giving the working precision dps, the script that produced the result, and the SHA-256 hash of the script source. This ledger forms part of the AEAL (Agent Epistemic Accountability Layer) protocol [2].

Manuscript preparation involved GitHub Copilot (OpenAI) and Anthropic Claude (Sonnet) in an advisory role: the agents drafted prose, suggested structural edits, and ran computations under the SIARC pipeline, but every mathematical statement was independently re-derived and validated by the author.

Reproducibility. The full source archive of scripts and certificates is hosted at https://github.com/papanokechi/siarc-relay-bridge, with key bridge commits `84f91bf` (degree-$(4,2)$ search), `fa259b0` (deep validation), `45fe389` (basis identification), `3a5bb2b` (LMFDB lookup), and `ce25196` (CMAX$=2$ ratio analysis).

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